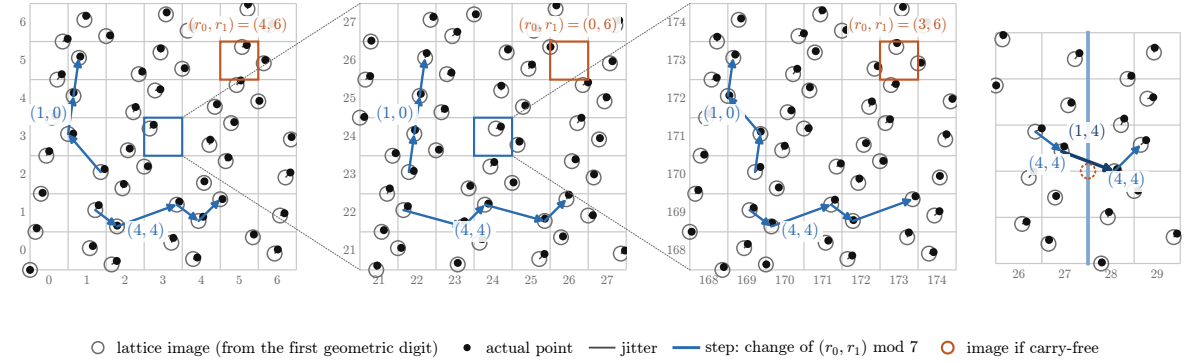
$K = 1$: P_2
side 7^{-1}; all steps carry length 1
$(r_0, r_1) = (4, 6)$
(1, 0)
(4, 4)
$K = 2$: P_4 in one level-1 cell
side 7^{-2}; all steps carry length 1
$(r_0, r_1) = (0, 6)$
(1, 0)
(4, 4)
$K = 3$: P_6 in one level-2 cell
side 7^{-3}; all steps carry length 1
$(r_0, r_1) = (3, 6)$
(1, 0)
(4, 4)
$K = 2$ across $n_0 = 28$:
carry length 2
(1, 4)
(4, 4)
(4, 4)
lattice image (from the first geometric digit)
actual point
jitter
step: change of (r_0, r_1) mod 7
image if carry-free